

ISO18000-6C Air Interface Protocol Function Library

V1.2

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1. Instruction

The development kit supports the ANSI-C, ANSI-C++ and other languages to call the C function interface, such as delphi, VB, C#. Currently, only 32-bit windows operating systems (32-bit window XP, 32-bit window 7) are supported. The development kit cannot be used independently and must be combined with other RFID development kit. Before using, you must read “DLL Development Guide.doc” first, to make sure well understand the whole development process for SDK package.

2. Application Integration

This package consists of the following files:

Files	Description
rfidlib_aip_iso18000p6C.dll	Dynamic link library, contains all API interface
rfidlib_aip_iso18000p6C.lib	VC++ project, static link library
rfidlib_aip_iso18000p6C.h	Header file of C/C++ project , the function definition
rfidlib_aip_iso18000p6C.cs	C# class file
rfidlib_aip_iso18000p6C.pas	Header file of Delphi function definition

Programming Language	Description
VC++	Copy rfidlib_aip_iso18000p6C.lib and rfidlib_aip_iso18000p6C.h to the project directory and add to the project. Copy rfidlib_aip_iso18000p6C.dll to the exe file output directory.
C#	Copy rfidlib_aip_iso18000p6C.cs to the project directory and add to the project. Copy rfidlib_aip_iso18000p6C.dll to the exe file output directory.
Delphi	Copy rfidlib_aip_iso18000p6C.pas to the project directory and add to the project. Copy rfidlib_aip_iso18000p6C.dll to the exe file output directory.

3. Programming Interface

3.1. Overview

The programming interface is on the basis of `rfidlib_reader.dll`, define standard protocol command interface for belongs to ISO18000-6C, make it convenient for application developers use EPC Gen2 protocol.

3.2. ISO18000p6C_GetLibVersion 【Get Library Version】

Definition	DWORD ISO18000p6C_GetLibVersion(LPTSTR buf ,DWORD nSize)		
Description	Get the version number of dynamic library		
Parameters	Name	Type	Remarks
	buf	LPTSTR	String buffer used for saving the version number
	nSize	DWORD	The max size of the string buffer
Return (DWORD)	Return the number of characters written to the buffer, if the return value is equal to “nSize”, shows that the version number length is greater than the max number of characters of the buffer		
Reference Code	<pre>char vbuf[128] ; DWORD written ; written = get_version(vbuf,128) ; if(written > 0) { /* show version string here */ printf(vbuf) ; } if(written == 128) { /* overflow,need to increase vbuf */ }</pre>		

3.3. ISO18000p6C_CreateInvenParam 【 Create Inventory Parameter】

Definition	RFID_DN_HANDLE RFIDLIB_API ISO18000p6C_CreateInvenParam(RFID_DN_HANDLE hInvenParamSpecList,BYTE AntennaID,BYTE Sel,BYTE Session,BYTE Target, BYTE Q)				
Description	Create ISO18000-6C Inventory Parameters data nodes, used for RDR_TagInventory function				
Parameters	Name	Type	Direction	Remarks	
	hInvenParamSpecList	RFID_DN_HANDLE	[in]	Inventory parameters list handle, created by RDR_CreateInvenParamSpecList.	
	AntennaID	BYTE	[in]	Antenna ID , 0 represents all antenna interfaces need to read tags at ISO18000-6C protocol , other value means corresponding antenna ports	
	Sel	BYTE	[in]	Select the tags to participate in the inventory: 0x00=inventory object is all tags 0x01=inventory object is all tags 0x02=inventory object without SL identification 0x03=inventory object with SL identification	
	Session	BYTE	[in]	Session type: 0x00: S0; 0x01: S1; 0x02:S2; 0x03:S3;	
	Target	BYTE	[in]	Target:	
				Value	Description
				0	Target A
				1	Target B
	Q	BYTE	[in]	Take the value range of 0-15, 0xFF is dynamic QValue	
Return	If successful, returns the handle of data node.				
Reference Code					

3.4. ISO18000p6C_SetInvenSelectParam 【Set Inventory Selection Parameter】

Definition	err_t ISO18000p6C_SetInvenSelectParam(RFID_DN_HANDLE hIso18000p6CInvenParam , BYTE target ,BYTE action ,BYTE memBank ,DWORD dwPointer,BYTE maskBits[], BYTE maskBitLen, BYTE truncate)			
Description	Enable ISO18000-6C inventory selection function			
Parameters	Name	Type	Direction	Remarks
	hIso18000p6CInvenParam	RFID_DN_HANDLE	[in]	The handle is created by ISO18000p6C_CreateInvenParam
	target	BYTE	[in]	Select effective target content: 0x00: inventoried mark of S0; 0x01: inventoried mark of S1; 0x02: inventoried mark of S2; 0x03: inventoried mark of S3; 0x04: SL
	action	BYTE	[in]	Actions to be taken on tags that meet the conditions: 0x00: Match: Insert SL or change inventoried→A ; Not match: Remove SL or change inventoried→B ; 0x01: Match: Insert SL or change inventoried→A ; Not match: Don't do anything ; 0x02: Match: Don't do anything ; Not match: Remove SL or change inventoried→B 0x03: Match: Invert SL or (A→B, B→A) Not match: Don't do anything ; 0x04: Match: Remove SL or change inventoried→B Not match: Insert SL or change inventoried→A 0x05: Match: Remove SL or change inventoried→B Not match: Don't do anything ;

				0x06: Match: Don't do anything ; Not match: Insert SL or change inventoried→A 0x07: Match: Don't do anything ; Not match: Invert SL or (A→B, B→A)
	memBank	BYTE	[in]	Storage region: 0x00: RFU; 0x01: EPC; 0x02:TID; 0x03:USER;
	dwPointer	DWORD	[in]	Match start bit pointer
	maskBits	BYTE[]	[in]	Matched bit data
	maskBitLen,	BYTE	[in]	Bit data length
	truncate	BYTE	[in]	Incoming 0, reserved, unused
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.5. ISO18000p6C_SetInvenMetaDataFlags 【Set Meta Flag】

Definition	err_t ISO18000p6C_SetInvenMetaDataFlags(RFID_DN_HANDLEhIso18000p6CInvenParam, DWORD flags)				
Description	Enable Meta flags, which determines the data fields of the tag records that inventory				
Parameters	Name	Type	Direction	Remarks	
	hIso18000p6CInvenParam	RFID_DN_HANDLE	[in]	This handle is created by ISO18000p6C_CreateInvenParam	
	Flags	DWORD	[in]	Meta flags bit:	
				Bit	Description
				b0	The returned record should include EPC fields
				b1	The returned record should include timestamp field
				b2	The returned record should include frequency field
				b3	The returned record should contain RSSI fields
				b4	The returned record should include the number of reads field
				b5	The returned record should contain data for the read operation.
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.6. ISO18000p6C_SetInvenReadParam 【 Set Inventory Reading Parameter】

Definition	err_t ISO18000p6C_SetInvenReadParam(RFID_DN_HANDLE hIso18000p6CInvenParam ,BYTE MemBank ,DWORD WordPtr,BYTE WordCount)			
Description	Enable inventory reading function			
Parameters	Name	Type	Direction	Remarks
	hIso18000p6CInvenParam	RFID_DN_HANDLE	[in]	This handle is created by ISO18000p6C_CreateInvenParam
	MemBank	BYTE	[in]	Storage area: 0x00: RFU; 0x01: EPC; 0x02:TID; 0x03:USER;
	WordPtr	DWORD	[in]	Pointer to the start word, a word=two bytes
	WordCount	BYTE	[in]	Count of word to read
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.7. ISO18000p6C_SetInvenAccessPassword【Set Access Password】

Definition	err_t ISO18000p6C_SetInvenAccessPassword(RFID_DN_HANDLE hIso18000p6CInvenParam ,DWORD pwd) ;			
Description	Set Read/Write Password			
Parameters	Name	Type	Direction	Remarks
	hIso18000p6CInvenParam	RFID_DN_HANDLE	[in]	This handle is created by ISO18000p6C_CreateInvenParam
	pwd	DWORD	[in]	Password
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.8. ISO18000p6_ParseTagDataReport 【 Parsing Tag Inventory Report】

Definition	err_t ISO18000p6C_ParseTagReport(RFID_DN_HANDLE hTagReport,DWORD *aip_id,DWORD *tag_id, DWORD * ant_id,DWORD *metaFlags,BYTE tagdata[],DWORD *tdLen)				
Description	Parsing ISO18000-6C tag records				
Parameters	Name	Type	Direction	Remarks	
	hTagReport	RFID_DN_HANDLE	[in]	Node handle of tag data information	
	aip_id	DWORD *	[out]	Return air interface protocol type ID, refer to type ID list of air interface protocol of “DLL Development Guide.doc”	
	tag_id	DWORD *	[out]	Return tag type ID, refer to Type ID list of tag of “DLL Development Guide.doc”	
	ant_id	DWORD *	[out]	Return antenna interface ID	
	metaFlags	DWORD *	[out]	Meta identifier, the value of each bit indicates whether the corresponding field exists in the “tagdata” field	
				Bit	Description
				b0	tagdata contains EPC fields
				b1	tagdata contains timestamp fields
				b2	tagdata contains frequency fields
				b3	tagdata contains RSSI fields
				b4	tagdata contains the number of reads field
				b5	tagdata contains data of the read operation
	tagdata	BYTE[]	[out]	Tag record data:	
					Field
1				EPC bit length (WORD)+ EPC data (BYTE []) Only when b0 of Meta is 1, this field exists;	
2				Timestamp (DWORD) Only when b1 of Meta is 1, this field exists;	
3				Frequency (DWORD) Only when b2 of Meta is 1, this	

					field exists;
				4	RSSI(BYTE) Only when b3 of Meta is 1, this field exists;
				5	Number of reads (BYTE) Only when b4 of Meta is 1, this field exists;
				6	The data of the read operation (BYTE[]), the length of the read data is the data after remove other data. Only when b5 of Meta is 1, this field exists.
	tdLen	DWORD *	[in] [out]	[in]: represents the maximum buffer size for saving tag data; [out]: represents the actual length of the obtained tag.	
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.9. ISO18000p6_ParseTagDataReportV2 【 Parsing Tag Inventory Report 】

Definition	err_t RFIDLIB_API ISO18000p6C_ParseTagReportV2(RFID_DN_HANDLE hTagReport,DWORD *aip_id,DWORD *tag_id,DWORD * ant_id,BYTE epc[],DWORD *dLen)			
Description	Parsing ISO18000-6C tag records and get the EPC of the tag			
Parameters	Name	Type	Direction	Remarks
	hTagReport	RFID_DN_HANDLE	[in]	Node handle of tag data information
	aip_id	DWORD *	[out]	Return air interface protocol type ID, refer to type ID list of air interface protocol of “DLL Development Guide.doc”
	tag_id	DWORD *	[out]	Return tag type ID, refer to Type ID list of tag of “DLL Development Guide.doc”
	ant_id	DWORD *	[out]	Return antenna interface ID
	epc	BYTE[]	[out]	EPC data for tags. Among them, epc [0] and epc [1] are the CRC values of the tag, while epc [2] and epc [3] represent the PC values of the tag.
	tdLen	DWORD *	[in] [out]	[in] represents the size of the maximum buffer for saving EPC; [out] represents the actual length of the obtained EPC.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.10. ISO18000p6C_CreateTAWrite 【Embedding Write Operation】

Definition	RFID_DN_HANDLE ISO18000p6C_CreateTAWrite(RFID_DN_HANDLE hIso18000p6CInvenParam, BYTE memBank, DWORD wordPtr ,DWORD wordCnt ,BYTE pdatas[],DWORD nSize) ;			
Description	Create embedded write operation			
Parameters	Name	Type	Direction	Remarks
	hIso18000p6CInvenParam	RFID_DN_HANDLE	[in]	This handle is created by ISO18000p6C_CreateInvenParam
	MemBank	BYTE	[in]	Storage area: 0x00: RFU; 0x01: EPC; 0x02:TID; 0x03:USER;
	WordPtr	DWORD	[in]	Pointer of the start word, a word=two bytes
	WordCnt	DWORD	[in]	Count of word to write
	pdatas	BYTE[]	[in]	Data
	nSize	DWORD	[in]	The length of data
Return	Success: returned node handle for write operation; Failure: returned 0			
Reference Code				

3.11. ISO18000p6C_CheckTAWriteResult 【 Parsing Result of Embedded Write】

Definition	err_t ISO18000p6C_CheckTAWriteResult(RFID_DN_HANDLE hTagReport)			
Description	Parsing the results of embedded write operations			
Parameters	Name	Type	Direction	Remarks
	hTagReport	RFID_DN_HAN DLE	[in]	Node handle for tag data information
Return	0: indicates successful writing of the corresponding tag; Non 0: indicates write failure			
Reference Code				

3.12. ISO18000p6C_CreateTALock 【Embedding Lock Operation】

Definition	RFID_DN_HANDLE RFIDLIB_API ISO18000p6C_CreateTALock(RFID_DN_HANDLE hIso18000p6CInvenParam, WORD mask ,WORD action)			
Description	Create embedded lock operation			
Parameters	Name	Type	Direction	Remarks
	hIso18000p6CInvenParam	RFID_DN_HANDLE	[in]	This handle is created by ISO18000p6C_CreateInvenParam
	mask	WORD	[in]	
	action	WORD	[in]	
Return	Success: returns the node handle for the lock operation; Failure: returns 0			
Reference Code				

Bit correspondence table of mask and action :

Bit	15-10	9	8	7	6	5	4	3	2	1	0
		Kill Pwd		Access Pwd		EPC Mem		TID Mem		User Mem	
Mask		1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0
Action		R/W	Perm	R/W	Perm	W	Perm	W	Perm	W	Perm

3.13. ISO18000p6C_Connect 【Connect Tag】

Definition	err_t RFIDLIB_API ISO18000p6C_Connect(RFID_READER_HANDLE hr,DWORD tag_id, BYTE epc[], BYTE epcLen,DWORD accessPwb, RFID_TAG_HANDLE *ht)			
Description	Connect tags. After the tag operation is completed, call RDR_TagDisconnect to release memory			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Handle
	tag_id	DWORD	[in]	Tag ID, default value is 0
	epc	BYTE	[in]	EPC data of tags
	epcLen	BYTE	[in]	The length of tag EPC data
	accessPwb	DWORD	[in]	Access password, default value is 0
	ht	RFID_TAG_HANDLE	[out]	Tag operation handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.14. ISO18000p6C_Read 【Read Tag Memory Data】

Definition	err_t RFIDLIB_API ISO18000p6C_Read(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE memBank, DWORD wordStart, DWORD wordCnt, BYTE wordData[], DWORD *nSize)			
Description	Read tag memory data			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	memBank	BYTE	[in]	Storage area: 0x00: RFU; 0x01: EPC; 0x02: TID; 0x03: USER;
	wordStart	BYTE	[in]	Read the starting word of memory
	wordCnt	DWORD	[in]	Number of words read from memory
	wordData	BYTE	[out]	Buffer for saving memory data
	nSize	DWORD *	[in] [out]	[in] represents the maximum buffer size for storing memory data; [out] represents the size of the memory area actually read
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.15. ISO18000p6C_Write 【Write Tag Memory Data】

Definition	err_t RFIDLIB_API ISO18000p6C_Write(RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht,BYTE memBank,DWORD wordStart,DWORD workCnt,BYTE wordData[],DWORD nSize)			
Description	Write tag memory data			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	memBank	BYTE	[in]	Storage area: 0x00: RFU; 0x01: EPC; 0x02:TID; 0x03:USER;
	wordStart	BYTE	[in]	Start word written to memory
	wordCnt	DWORD	[in]	The number of words written to memory
	wordData	BYTE	[in]	Data that needs to be written
	nSize	DWORD	[in]	[in] represents the size of the data written to memory
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.16. ISO18000p6C_Lock 【Lock Tag】

Definition	err_t RFIDLIB_API ISO18000p6C_Lock(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, WORD mask, WORD action)			
Description	Lock tag			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	mask	WORD	[in]	Reference 3.12
	action	WORD	[in]	Reference 3.12
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.17. ISO18000p6C_Kill 【Destroy Tag】

Definition	err_t RFIDLIB_API ISO18000p6C_Kill(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, DWORD killPwb, BYTE rfu)			
Description	Destroy tag			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	killPwb	DWORD	[in]	Password required to destroy the tag
	rfu	BYTE	[in]	Reserved data, default value is 0
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				